

ODEs

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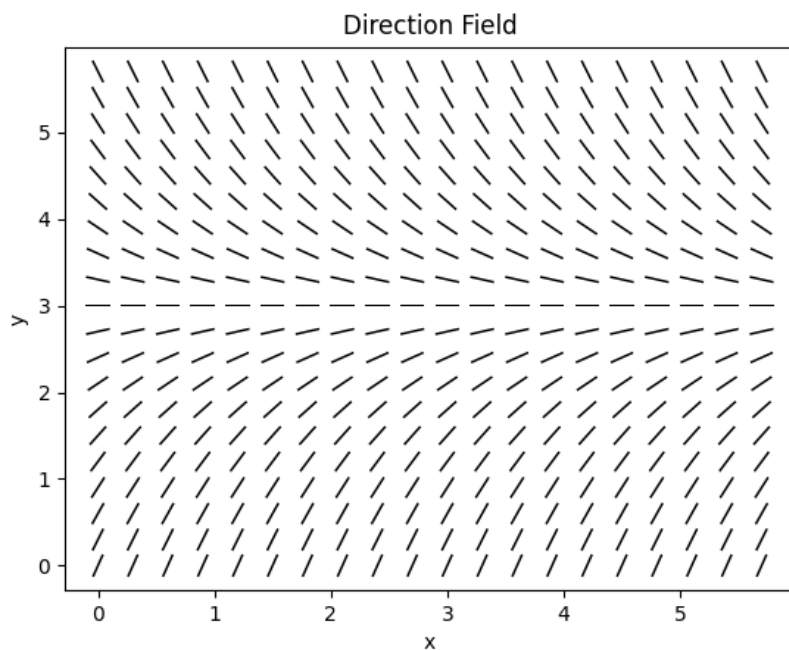
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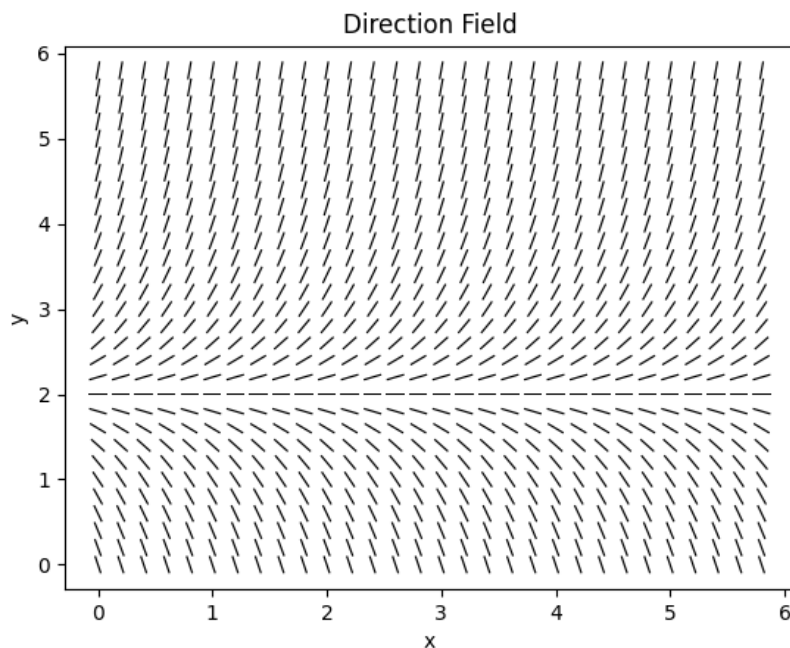
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1 Section 1.1

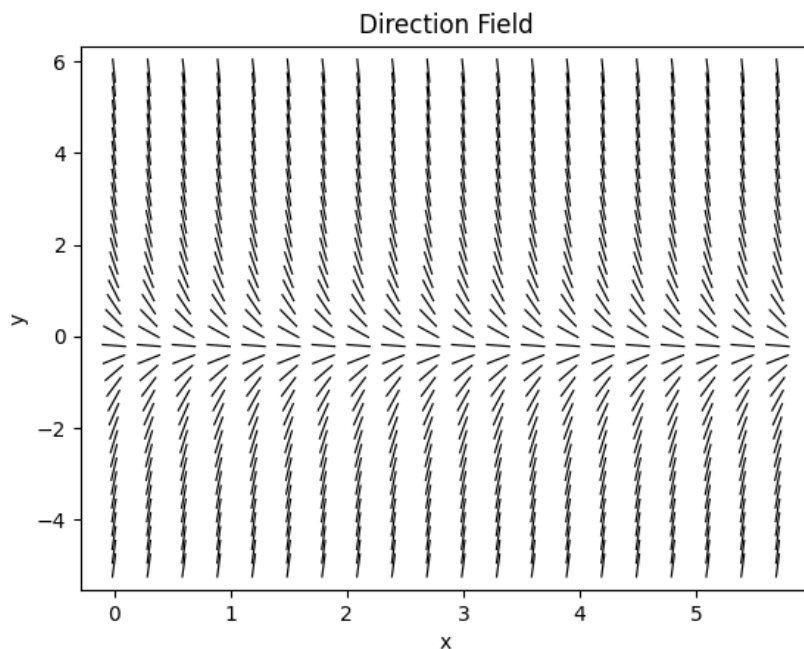
Problems 1, 2, 3, 4



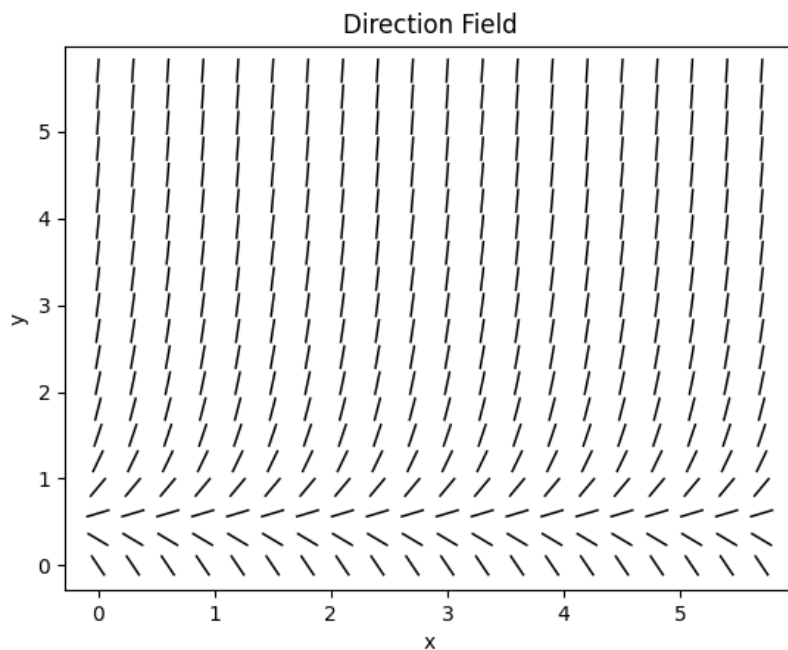
Problem 1: Above shows the direction field for $y' = 3 - y$. It can be seen that for $y > 3$ the slopes are negative so solutions decrease and for $y < 3$ the slopes are positive so solutions increase. The equilibrium position appears to be at $y(t) = 3$, to which all other solutions converge.



Problem 2: Above shows the direction field for $y' = 2y - 4$. For $y > 2$ the slopes are positive so solutions increase and for $y < 2$ the slopes are negative so solutions decrease. The equilibrium solution appears to be $y = 2$ and all other solutions diverge away from this solution.



Problem 3: Above shows the direction field for $y' = -1 - 4y$. For $y > -\frac{1}{4}$ the slopes are positive so solutions increase. For $y < -\frac{1}{4}$ the slopes are negative so solutions decrease. The equilibrium solution appears to be $y = -\frac{1}{4}$ and all other solutions converge to this solution.



Problem 4: Above shows the direction field for $y' = 4y - 2$. For $y > 0.5$ the slopes are positive so solutions are increasing. For $y < 0.5$ the slopes are negative so the solutions are decreasing. It appears that the equilibrium solution is at $y = \frac{1}{2}$ and all other solutions diverge away.

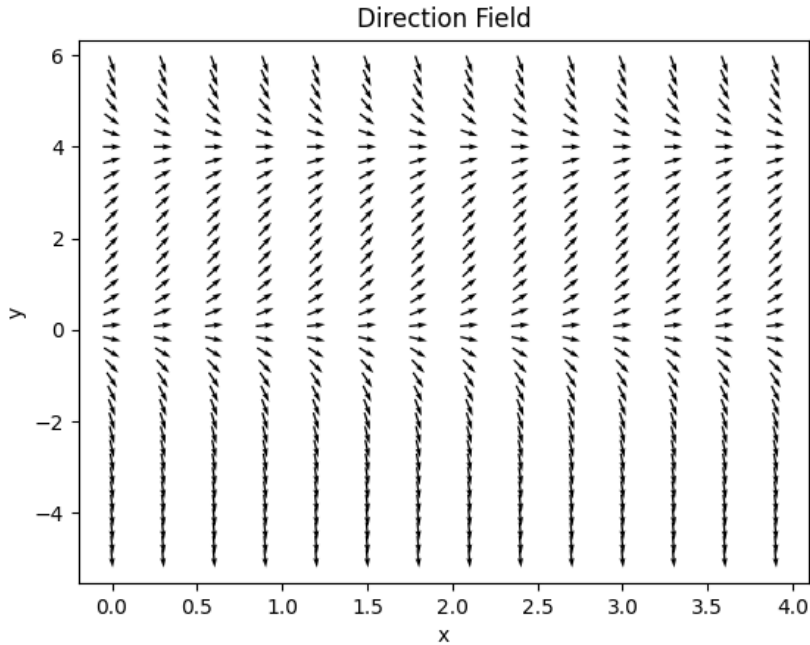
Problems 5 and 6

Problem 5: A differential equation where all solutions approach $y = \frac{2}{3}$ is given by $\frac{dy}{dx} = 2 - 3y$.

Problem 6: A differential equations where all other solutions diverge from $y = 3$ is $\frac{dy}{dx} = y - 3$

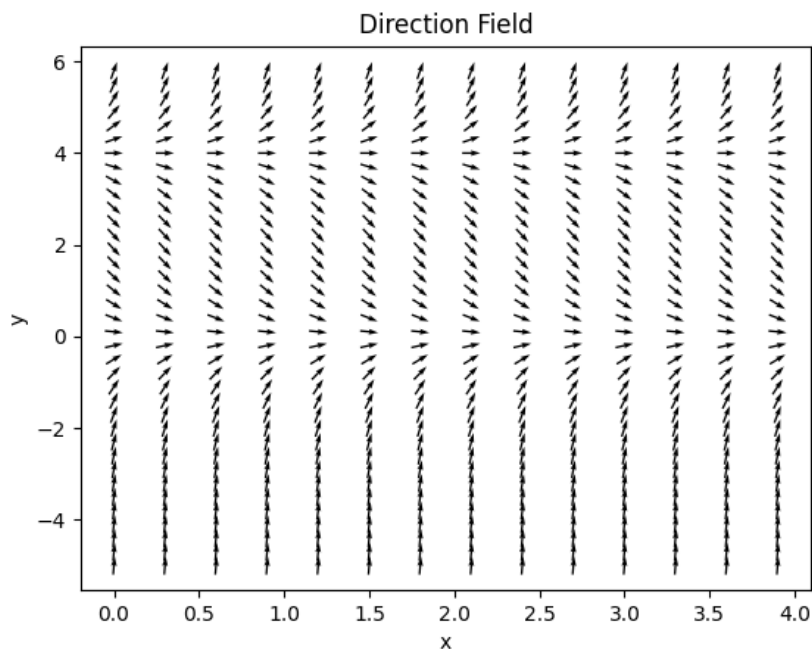
Problems 7, 8, 9, 10

Problem 7:



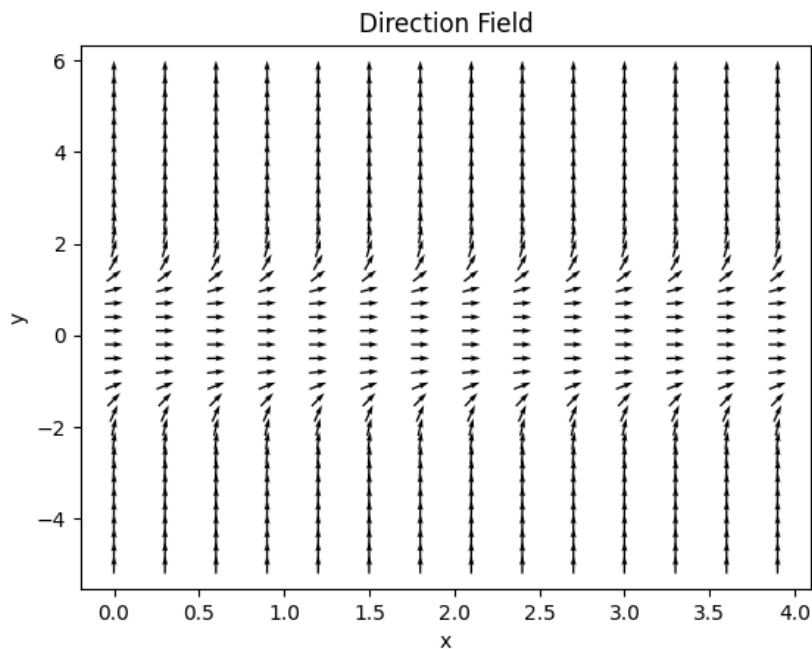
For $y > 4$, the slopes are negative and so solutions decrease. For $0 < y < 4$, slopes are positive so solutions increase. For $y < 0$ slopes are negative so solutions decrease. Equilibrium solutions appear to be at $y = 4$ to which solutions converge to and $y = 0$ at which solutions diverge from.

Problem 8:



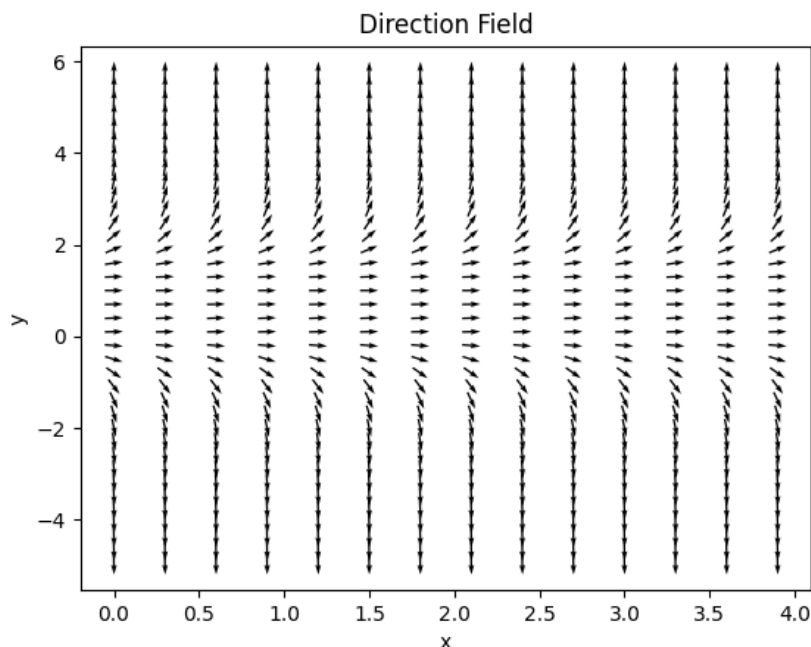
For $y > 4$, the slopes are positive and so solutions increase. For $0 < y < 4$, slopes are negative so solutions decrease. For $y < 0$ slopes are positive so solutions decrease. Equilibrium solutions appear to be at $y = 0$ to which solutions converge to and $y = 4$ at which solutions diverge from.

Problem 9:



For $y > 0$, the slopes are positive and so solutions increase. For $y < 0$, the slopes are negative and so solutions increase. There appears to be an equilibrium position at $y = 0$ where solutions greater than 0 diverge from and solutions less than 0 converge to.

Problem 10:



For $y > 1$, the slopes are positive and so solutions are increasing For $0 < y < 1$, the slopes are positive so solutions are increasing For $y < 0$, the slopes are negative and so solutions are decreasing Equilibrium solutions appear to be at $y = 0$ which is unstable and $y = 1$ which is semi-stable.

Problems 11, 12, 13, 14, 15, 16

Problem 11 The direction field of **Figure 1.1.5** has an equilibrium solution at $y = 2$ to which all solutions converge to. So the differential equation that corresponds to this direction field is $\frac{dy}{dx} = 2 - y$.

Problem 12 The direction field of **Figure 1.1.6** has an equilibrium solution at $y = 2$ which is unstable. So the differential equation that corresponds to this direction field is $\frac{dy}{dx} = y + 2$.

Problem 13 The direction field of **Figure 1.1.7** has an equilibrium solution at $y = -3$ which is stable. So the differential equation that corresponds to this direction field is $\frac{dy}{dx} = -3 - y$.

Problem 14 The direction field of **Figure 1.1.8** has an equilibrium solution at $y = -1$ which is unstable. So the differential equation that corresponds to this direction field is $\frac{dy}{dx} = y + 1$.

Problem 15 The direction field of **Figure 1.1.9** has an equilibrium solution at $y = 2$ which is stable and $y = 0$ which is unstable. So the differential equation that corresponds to this direction field is $\frac{dy}{dx} = y(2 - y)$.

Problem 16 The direction field of **Figure 1.1.10** has an equilibrium solution at $y = 0$ which is stable and $y = 3$ which is unstable. So the differential equation that corresponds to this direction field is $\frac{dy}{dx} = y(y - 3)$.